

P = pre-mutation matrix

- Shows which rows are swapped
- Initially a copy of base L
- Row swap also changes sign of determinant

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ from Base L}$$

Swap rows 1 & 3 on A:

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ Result of partial pivot}$$

L = lower triangular matrix

- Trades factors & manipulations
- Row = unmanipulated row
- Column = manipulated row
- Value = factor by which column is scaled

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ Base; factors replace zeros}$$

If manipulation $R_3 = R_3 - 2R_2$ performed: $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

unmanipulated ← manipulated
manipulation factor

U = upper triangular matrix

- Multiply by e for answers
- Manipulate A so it has zeroes below diagonal

$$U = \begin{bmatrix} c_1 & c_2 & c_3 \\ 0 & c_4 & c_5 \\ 0 & 0 & c_6 \end{bmatrix}$$

PA = LU

- PA is "A w/ swaps of P"
- LU is "reverse-engineering U w/ L"
- Row transforms are independent of each other; top → bottom, left → right

Gaussian Elimination:

- ① **Partial pivoting:** Largest value in 1st column should also be in 1st row (P)
- ② **Row manipulation:** bottom left below diagonal should be zeroes (L)
- ③ **Determinant:** Multiply all values of U diagonal by partial pivot coefficient (|A|)
- ④ **Verify PA = LU:** Apply P swaps on A & apply reverse of L factors on U until they're equal
- ⑤ **Determinant of Inverse:** $|A^{-1}| = \frac{1}{|A|}$

I = identity matrix

- 1 on main diagonal, 0 everywhere else
- P = I if no rows are flipped
- PLUA⁻¹ = I if P ≠ I; otherwise LUA⁻¹ = I

$$I = \begin{bmatrix} 1 & 0 & 0 & \vdots \\ 0 & 1 & 0 & \vdots \\ 0 & 0 & 1 & \vdots \\ \vdots & \vdots & \vdots & 1 \end{bmatrix}$$

Ax = e_j

- jth column of A is solution x for Ax = e_j
- e_j is vector w/ 1 in position j, 0 elsewhere
- Ly = e_j & Ux = y

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ for 3 rows in A}$$

$$Ly = \begin{bmatrix} n_1 & n_2 & n_3 \\ n_4 & n_5 & n_6 \\ n_7 & n_8 & n_9 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = e_j \quad U \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

A = LU

- Fake dot product to find A

$$A_{(i,j)} = \begin{bmatrix} m_1 & m_2 & m_3 \\ m & m & m \\ m & m & m \end{bmatrix} \begin{bmatrix} n_1 & n & n \\ n_4 & n & n \\ n_7 & n & n \end{bmatrix} = m_1n_1 + m_2n_4 + m_3n_7$$

Norms

- ||A|| is matrix norm
- ||A||₁ is 1-norm; max column sum of absolute values
- ||A||_∞ is ∞-norm; max row sum of absolute values
- κ(A) is condition number; κ(A) = ||A|| ||A⁻¹||
- ||A|| ≤ √(||A||₁ ||A||_∞); also applies to ||A⁻¹||

LU Decomposition:

- ① **Find Missing Column:** Use Ly = e_j to find y vector, sub in to Ux = y to solve for x vector, plug into missing column
- ② **Find Condition #:** Find 1-norm & ∞-norm for ||A|| & ||A⁻¹||, & use to find norms, then κ(A)

Bisection Method

- Check midpoint $c = \frac{a+b}{2}$ until f(c) ≈ 0

n	a	b	f(a)	f(b)	c	f(c)
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- f(c) replaces whichever value has same sign

False Position Method

$$c = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

n	a	b	f(a)	f(b)	c	f(c)
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$$f(c) = \frac{f(b) - f(a)}{b - a} (c - a) + f(a)$$

- f(c) replaces whichever value has same sign

Newton Method

- Assume starting point (x₀, f(x₀))
- Check f'(x₀)
- Next guess is intersection b/t f'(x₀) & x-axis
- For n iterations, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
- Has to fail if you use it first

Taylor Series

- $\log(x) \approx (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \dots$
- $\frac{1}{\sqrt{x}} \approx 1 - \frac{1}{2}(x-1) + \frac{3}{8}(x-1)^2 - \frac{5}{16}(x-1)^3 + \dots$
- $e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$
- $\sin(x) \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$
- $\cos(x) \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

Horner's Algorithm:

- ① **Find Taylor series:** $T(x) = \sum_{n=0}^k \frac{f^{(n)}(c)}{n!} (x-c)^n$
- ② **Do synthetic division:** Divide by given x subtracted by where you center T(x)

Kull-Dobell Theorem to Maximize Period

- Modulus m, multiplier a, increment c, seed x₀
- c cannot be divisible by any of m's prime factors
- a-1 must be divisible by all of m's prime factors
- If m is divisible by 4, so is a-1
- $(a * x_n + c) \% m = x_{n+1}$

Random Number Generators:

- ① **Find m prime factors:** Use to find lowest possible # of c
- ② **Find a:** Can set a-1=m for ease
- ③ **Randomness:** Easier to spot patterns if results are in binary

1. (Binary Floating Point) Assume an 8-bit floating point format with a 4-bit exponent and a bias of 7 and implied 1.

- Provide the decimal representations of the following stored values.

Sign	Exponent	Mantissa
1	0 0 1 1	0 1 0 1 0
-1	3 ₁₀	2 ₁₀

Value = $(-1)^{\text{sign}} \times (1 + \frac{\text{mantissa}}{2^4}) \times 2^{\text{exp.} - \text{bias}}$

→ $-1 \times 2^{3-7} \times (1 + \frac{10}{16}) = -0.078125$

Sign	Exponent	Mantissa
0	1 0 1 1	1 1 1 0
+1	10 ₁₀	6 ₁₀

→ $1 \times 2^{10-7} \times (1 + \frac{6}{8}) = 14$

- For this format, if the spacings between any two adjacent floating points are the same? If your answer is yes, please provide this spacing in decimal value. If your answer is no, please provide the largest and smallest spacings in decimal values.

No. The spacing depends on the exponent in this format.

#s w/ same exponents & adjacent mantissas have equal spacing.

$$E_{\min} = 0-7 = -7 \quad E_{\max} = 15-8 = 7 \quad \Delta_{\min} = 2^{-7-3} = \frac{1}{1024} \quad \Delta_{\max} = 2^{8-3} = 32$$

2. (Horner's Algorithm) Calculate $f(x) = \frac{\cos(x-1)}{\sqrt{x}}$ for $x = 1.2$ using Taylor Series Expansion and Horner's algorithm with an error of $O(x^4)$ and 3 digits of precision. Demonstrate Horner's algorithm using synthetic division. Show your work, including all derivations.

$$\cos(x-1) \approx 1 - \frac{(x-1)^2}{2!} + \frac{(x-1)^4}{4!} - \dots \rightarrow \cos(x-1) \approx 1 - \frac{(x-1)^2}{2} + O(x^4)$$

$$\frac{1}{\sqrt{x}} = x^{-1/2} \approx 1 - \frac{1}{2}(x-1) + \frac{3}{2!}(x-1)^2 + O(x^4)$$

$$f(x) \approx 1 - \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2 - \frac{1}{6}(x-1)^3 + O(x^4) = \left(-\frac{1}{6}(x-1) - \frac{1}{8}(x-1) - \frac{1}{2}\right)(x-1) + 1$$

$$(x-1) = 0.2 \quad \begin{array}{r} -\frac{1}{6} \quad -\frac{1}{8} \quad -\frac{1}{2} \quad 1 \\ \hline -\frac{1}{6} \quad -\frac{11}{24} \quad -\frac{24}{240} \\ -\frac{11}{24} \quad -\frac{24}{240} \quad -\frac{1789}{2400} \end{array} \approx 0.895$$

10. (15 pts) Use synthetic division to compute $f(x) = e^{-x} \sin x^2$ for $x = 0.1$ with an error bound of $O(x^4)$ using 3 significant digits. Show this calculation using synthetic division.

$$e^{\wedge}x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$$

$$e^{\wedge}x = 1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3$$

$$\sin x = \sin(0) + \cos(0)x$$

$$\sin x^2 = x^2$$

$$f(x) = x^2 - x^3$$

0.1	-1	1	0	0
	-0.1	0.09	0.00900	
	-1	0.9	0.09	0.00900

3. Compare the relative errors for the following integrals using two iterations (intervals) of trapezoid rule and one iteration (interval) of Simpson's rule. Use three digits of precision where appropriate.

- i. (P12.1)

$$\int_0^1 x(x-1)(x-2) dx \quad \text{Exact: } 0.25$$

Trapezoid: $\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + 2f(x) + f(b)] \quad h = \frac{b-a}{n} = \frac{1-0}{2} = \frac{1}{2} = 0.5$

$$\int_0^1 x(x-1)(x-2) dx \approx \frac{0.5}{2} [f(0) + 2f(0.5) + f(1)] = \frac{0.5}{2} [0 + 2(0.375) + 0] = 0.25(0.75) = 0.188$$

$$\% \text{ error} = \frac{|0.188 - 0.25|}{0.25} \times 100\% = 24.8\%$$

Simpson: $\int_a^b f(x) dx \approx \frac{b-a}{6} [f(a) + 4f(x) + f(b)] \quad h = \frac{b-a}{n} = \frac{1-0}{2} = 0.5$

$$\int_0^1 x(x-1)(x-2) dx \approx \frac{0.5}{6} [f(0) + 4f(0.5) + f(1)] = \frac{1}{6} [0 + 4(0.375) + 0] = \frac{1}{6} (1.5) = 0.25 \quad \% \text{ error} = 0\%$$

One iteration of Simpson's rule gives no error, while 2 iterations of trapezoid rule gives significant error.

- ii. (P12.2)

$$\int_0^1 \sqrt{1-x^4} dx = 0.874 \quad h = 0.5$$

$$\int_0^1 \sqrt{1-x^4} dx \approx \frac{0.5}{2} [f(0) + 2f(0.5) + f(1)] = 0.25 [1 + 2(0.968) + 0] = 0.25(1.936) = 0.484$$

$$\% \text{ error} = \frac{|0.484 - 0.874|}{0.874} \times 100\% = 44.6\%$$

$$\int_0^1 \sqrt{1-x^4} dx \approx \frac{0.5}{6} [f(0) + 4f(0.5) + f(1)] = \frac{1}{6} [1 + 4(0.968) + 0] = \frac{1}{6} (4.872) = 0.812$$

$$\% \text{ error} = \frac{|0.812 - 0.874|}{0.874} \times 100\% = 7.09\%$$

Simpson's rule gives lower error than trapezoid rule.